

Day 3 : RAM and Storage Theory

Task 1 : Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

DDR3 vs DDR4 vs DDR5 RAM

Feature	DDR3	DDR4	DDR5
Data Speed	800–2133 MT/s	1600–3200 MT/s	4800 MT/s and higher
Voltage	About 1.5V	About 1.2V	About 1.1V
Performance	Lower	Better	Highest
Power Use	Higher	Lower than DDR3	Lower than DDR4
Typical Use	Older PCs and laptops	Many mainstream PCs and laptops	Modern PCs, gaming PCs, and high-performance systems
Availability	Older/legacy	Common but being replaced	Current/newer systems

Simple Explanation

DDR3: Older RAM technology. It is mainly found in older computers and is slower than DDR4 and DDR5.

DDR4: Faster and more power-efficient than DDR3. It has been widely used in desktops and laptops for many years.

DDR5: The newest of the three and offers higher speed and better performance. It is commonly used in newer gaming PCs and modern computers.

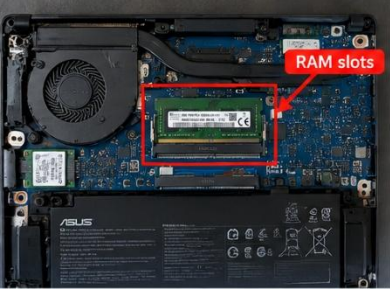
Remember:

DDR3 → Older & slower

DDR4 → Good performance

DDR5 → Newer & faster

Task 2 : Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.



1 Laptop opened and RAM slots located (after removing back cover and disconnecting battery).



2 Close-up photo of the installed RAM module (SO-DIMM DDR4).



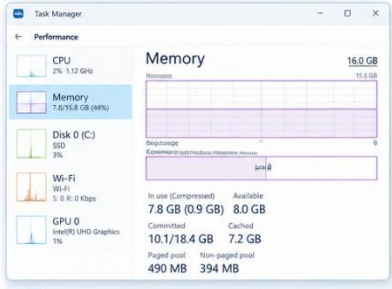
3 RAM module removed by pressing the clips on both sides.



4 RAM module reinstalled and clips locked into place.



5 Laptop reassembled and booted successfully.



6 System recognizes the RAM (checked in Task Manager → Performance → Memory).

✓ Result: The system booted successfully and the RAM was detected.

RAM Removal and Reinstallation

- 1. Locate RAM:** The laptop is opened with the power disconnected, and the RAM slot is located.
- 2. Remove RAM:** The RAM module is carefully removed by releasing the side clips.
- 3. Take Photo:** A clear image of the RAM module and RAM slot is taken.
- 4. Reinstall RAM:** The RAM module is placed back into the slot and pressed until the clips lock.

5. Boot the Laptop: The laptop is reassembled and powered on.

6. Verify RAM: Windows Task Manager → **Performance** → **Memory** is checked to confirm that the RAM is detected.

Result: The system boots successfully and the RAM is recognized.

Task 3 : List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

HDD vs SSD vs NVMe

Feature	HDD	SSD	NVMe SSD
Speed	Slow	Fast	Very fast
How it works	Uses spinning disks	Uses flash memory	Uses flash memory + PCIe
Moving Parts	Yes	No	No
Best Use	Large file storage	Daily apps and gaming	Heavy apps and large files
Daily Use Example	Photo/video backup	Spotify, Windows	Video editing or high-end gaming

1. HDD

HDD (Hard Disk Drive) stores data on spinning magnetic disks. It is slower than SSD and NVMe, but it is usually cheaper and available in large capacities.

Example: Storing movies, photos, documents, and backups.

2. SSD

SSD (Solid State Drive) uses flash memory and has no moving parts, so it is faster and more durable than an HDD. It is a good choice for Windows, everyday applications, and gaming.

Example: Spotify can load faster when installed on an SSD.

3. NVMe SSD

NVMe SSD is a high-speed type of SSD that uses the PCIe interface. It is much faster for large file transfers and demanding applications.

Example: Adobe Premiere Pro benefits from NVMe when working with large video files.

Easy to remember:

HDD → Large & Cheap

SSD → Fast & Common

NVMe → Fastest & High Performance

Task 4 : Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

ASUS TUF Gaming F15 – Remove and Reinstall Storage Drive (M.2 NVMe SSD)

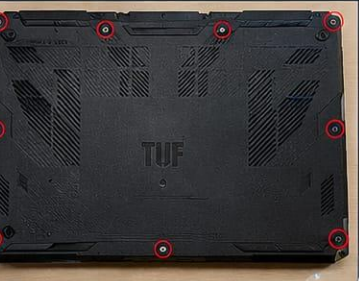
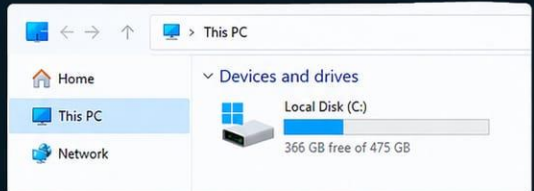
- 1 Open the laptop**
Turn off the laptop, remove the power cable, then unscrew the bottom cover (as shown).

- 2 Locate the M.2 slot**
After removing the bottom cover, you'll see the M.2 SSD slot (usually under a small screw and a heat shield).

- 3 Remove the SSD**
Unscrew the retaining screw and gently pull the SSD out at an angle.

- 4 Connector type (M.2 NVMe)**
The ASUS TUF Gaming F15 uses an M.2 NVMe SSD. The connector looks like this:

- 5 Reinstall the SSD**
Insert the SSD back into the slot at the same angle, then press it down and secure the retaining screw. Reattach the heat shield (if there is one).

- 6 Close the laptop**
Put the bottom cover back on and screw it in place.

- 7 Confirm the device recognizes the storage**
Boot into Windows and check in "This PC" or "Disk Management".

**Result**
The storage drive is detected and working properly!

Summary for ASUS TUF Gaming F15

- **Connector type:** M.2 NVMe (PCIe)
- **Location:** Inside (under bottom cover, near the cooling fan)
- **Steps:** Remove bottom cover → Unplug SSD → Reinstall → Close → Verify in Windows

Note: The exact screw locations and layout may vary slightly depending on the specific model (2021/2022/2023/2024/2025).

Remove and Reinstall Storage Drive

Device: ASUS TUF Gaming F15

Storage Type: M.2 NVMe SSD

Connector Type: M.2 NVMe (PCIe)

Procedure

1. **Turn off the laptop** and disconnect the power cable.
2. **Remove the bottom cover** of the ASUS TUF Gaming F15.
3. **Locate the M.2 NVMe SSD** inside the laptop.
4. **Remove the retaining screw** and gently remove the SSD from the M.2 slot.
5. **Check the connector:** The SSD uses an M.2 NVMe (PCIe) connector.
6. **Reinstall the SSD** into the M.2 slot at an angle and secure it with the screw.
7. **Close the bottom cover** and screw it back into place.
8. **Start Windows** and check This PC or Disk Management.
9. **Result:** The storage drive is detected and working properly.

Conclusion

The ASUS TUF Gaming F15 uses an M.2 NVMe SSD with an M.2 PCIe connector. After reinstalling the SSD, the storage is recognized by Windows.